Introducción a MongoDB

Módulo 5

Implementación de MongoDB Sharding

Implementación del Sharding

A continuación se seguirán, desde aquí, los pasos para fragmentar.

Paso 1

Se inicia el **servidor de configuración** del conjunto de réplicas desde la consola y se habilita la réplica entre ellos.

```
mkdir \data\config\rs0 \data\config\rs1 \data\config\rs2

mongod --replSetcsReplSet --dbpath \data\config\rs0 --port57017 --configsvr
mongod --replSetcsReplSet --dbpath \data\config\rs1 --port57018 --configsvr
mongod --replSetcsReplSet --dbpath \data\config\rs2 --port57019 --configsvr
```

```
Cmder
λ mkdir \data\config\rs0 \data\config\rs1 \data\config\rs2

E:\
λ |

E:\
λ mongod --replSet csReplSet --dbpath \data\config\rs0 --port 57017 --configsvr
{"t":{"$date":"2022-05-22T11:37:17.686-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0
, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-22T11:37:17.687-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}

E:\
λ mongod --replSet csReplSet --dbpath \data\config\rs1 --port 57018 --configsvr
{"t":{"$date":"2022-05-22T11:37:23.984-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T11:37:23.985-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0
, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}

E:\
λ mongod --replSet csReplSet --dbpath \data\config\rs2 --port 57019 --configsvr
{"t":{"$date":"2022-05-22T11:37:37.511-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0
, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-22T11:37:37.514-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificati
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"max

cmd.exe   mongod.exe   mongod.exe   mongod.exe   Search
```

Paso 2

En una consola independiente, se incializa el conjunto de réplicas en uno de los servidores de configuración.

```
mongo --port57017

config = { _id:"csReplSet", members: [
    {_id:0, host:'localhost:57017'},
    {_id:1, host:'localhost:57018'},
    {_id:2, host:'localhost:57019'}
]};

rs.initiate(config)
rs.status()
```

```
Cmder

E:\
λ mongo --port 57017
MongoDB shell version v5.0.8
connecting to: mongodb://127.0.0.1:57017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("e794170f-1398-457e-88bd-9c644c3ffb0f") }
MongoDB server version: 5.0.8
================
Warning: the "mongo" shell has been superseded by "mongosh",
which delivers improved usability and compatibility.The "mongo" shell has been deprecated and will be removed in
an upcoming release.
For installation instructions, see
https://docs.mongodb.com/mongodb-shell/install/
================
---
The server generated these startup warnings when booting:
        2022-05-22T11:37:18.674-03:00: Access control is not enabled for the database. Read and write access to data and configuration
is unrestricted

E:\
λ mongod --replSet csReplSet --dbpath \data\config\rs0 --port 57017 --configsvr

E:\
λ mongod --replSet csReplSet --dbpath \data\config\rs1 --port 57018 --configsvr

E:\
λ mongod --replSet csReplSet --dbpath \data\config\rs2 --port 57019 --configsvr

mongo.exe   mongod.exe   mongod.exe   mongod.exe   Search
```

El resultado en consola sería el siguiente:

```
E:\
λ mongo --port 57017
MongoDB shell version v5.0.8
connecting to: mongodb://127.0.0.1:57017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("e794170f-1398-457e-88bd-9c644c3ffb0f") }
MongoDB server version: 5.0.8

> config = { _id:"csReplSet", members: [     {_id:0, host:'localhost:57017'},     {_id:1,
host:'localhost:57018'},     {_id:2, host:'localhost:57019'} ]};
{
"_id" : "csReplSet",
"members" : [
{
"_id" : 0,
"host" : "localhost:57017"
},
{
"_id" : 1,
"host" : "localhost:57018"
},
```

```
...
{
"_id" : 1,
"host" : "localhost:57018"
},
{
"_id" : 2,
"host" : "localhost:57019"
}
]
}
```

```
> rs.initiate(config)
{
"ok" : 1,
"$gleStats" : {
"lastOpTime" : Timestamp(1653230393, 1),
"electionId" : ObjectId("000000000000000000000000")
},
"lastCommittedOpTime" : Timestamp(1653230393, 1)
}
```

```
csReplSet:SECONDARY> rs.status()
{
"set" : "csReplSet",
"date" : ISODate("2022-05-22T14:40:03.270Z"),
"myState" : 2,
"term" : NumberLong(0),
"syncSourceHost" : "",
"syncSourceId" : -1,
"configsvr" : true,
"heartbeatIntervalMillis" : NumberLong(2000),
"majorityVoteCount" : 2,
"writeMajorityCount" : 2,
"votingMembersCount" : 3,
"writableVotingMembersCount" : 3,
"optimes" : {
"lastCommittedOpTime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
```

...

```
...
"lastCommittedWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"readConcernMajorityOpTime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"appliedOpTime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"durableOpTime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"lastAppliedWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:39:53.562Z")
},
...
```

```
"lastStableRecoveryTimestamp" : Timestamp(1653230393, 1),
"members" : [
{
"_id" : 0,
"name" : "localhost:57017",
"health" : 1,
"state" : 2,
"stateStr" : "SECONDARY",
"uptime" : 166,
"optime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
```

```
"optimeDate" : ISODate("2022-05-22T14:39:53Z"),
"lastAppliedWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"syncSourceHost" : "",
"syncSourceId" : -1,
"infoMessage" : "Could not find member to sync from",
"configVersion" : 1,
"configTerm" : 0,
"self" : true,
"lastHeartbeatMessage" : ""
},
```

```
{
"_id" : 1,
"name" : "localhost:57018",
"health" : 1,
"state" : 2,
"stateStr" : "SECONDARY",
"uptime" : 9,
"optime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
```

```
...
"optimeDurable" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"optimeDate" : ISODate("2022-05-22T14:39:53Z"),
"optimeDurableDate" : ISODate("2022-05-22T14:39:53Z"),
"lastAppliedWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"lastHeartbeat" : ISODate("2022-05-22T14:40:02.826Z"),
"lastHeartbeatRecv" : ISODate("2022-05-22T14:40:02.896Z"),
"pingMs" : NumberLong(0),
"lastHeartbeatMessage" : "",
"syncSourceHost" : "",
"syncSourceId" : -1,
"infoMessage" : "",
"configVersion" : 1,
"configTerm" : 0
},
...
```

```
{
"_id" : 2,
"name" : "localhost:57019",
"health" : 1,
"state" : 2,
"stateStr" : "SECONDARY",
"uptime" : 9,
"optime" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"optimeDurable" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
```

...

```
"optimeDate" : ISODate("2022-05-22T14:39:53Z"),
"optimeDurableDate" : ISODate("2022-05-22T14:39:53Z"),
"lastAppliedWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:39:53.562Z"),
"lastHeartbeat" : ISODate("2022-05-22T14:40:02.826Z"),
"lastHeartbeatRecv" : ISODate("2022-05-22T14:40:02.883Z"),
"pingMs" : NumberLong(0),
"lastHeartbeatMessage" : "",
"syncSourceHost" : "",
"syncSourceId" : -1,
"infoMessage" : "",
"configVersion" : 1,
"configTerm" : 0
}
],
"ok" : 1,
"$gleStats" : {
"lastOpTime" : Timestamp(1653230393, 1),
"electionId" : ObjectId("000000000000000000000000")
},
```

...

```
"lastCommittedOpTime" : Timestamp(1653230393, 1),
"$clusterTime" : {
"clusterTime" : Timestamp(1653230393, 1),
"signature" : {
"hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
"keyId" : NumberLong(0)
}
},
"operationTime" : Timestamp(1653230393, 1)
}

csReplSet:SECONDARY>
```

Luego de unos instantes, el primer servidor se vuelve primario. Se verifica el status del Replica Set del Config Server, como se muestra a continuación.

```
csReplSet:SECONDARY>
csReplSet:PRIMARY> rs.status()
{
"set" : "csReplSet",
"date" : ISODate("2022-05-22T14:45:25.077Z"),
"myState" : 1,
"term" : NumberLong(1),
"syncSourceHost" : "",
"syncSourceId" : -1,
"configsvr" : true,
"heartbeatIntervalMillis" : NumberLong(2000),
"majorityVoteCount" : 2,
"writeMajorityCount" : 2,
"votingMembersCount" : 3,
"writableVotingMembersCount" : 3,
"optimes" : {
"lastCommittedOpTime" : {
"ts" : Timestamp(1653230724, 1),
"t" : NumberLong(1)
},
```

```
"lastCommittedWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"readConcernMajorityOpTime" : {
"ts" : Timestamp(1653230724, 1),
"t" : NumberLong(1)
},
"appliedOpTime" : {
"ts" : Timestamp(1653230724, 1),
"t" : NumberLong(1)
},
"durableOpTime" : {
"ts" : Timestamp(1653230724, 1),
"t" : NumberLong(1)
},
"lastAppliedWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:45:24.926Z")
},
```

```
...
"lastStableRecoveryTimestamp" : Timestamp(1653230694, 1),
"electionCandidateMetrics" : {
"lastElectionReason" : "electionTimeout",
"lastElectionDate" : ISODate("2022-05-22T14:40:04.314Z"),
"electionTerm" : NumberLong(1),
"lastCommittedOpTimeAtElection" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"lastSeenOpTimeAtElection" : {
"ts" : Timestamp(1653230393, 1),
"t" : NumberLong(-1)
},
"numVotesNeeded" : 2,
"priorityAtElection" : 1,
"electionTimeoutMillis" : NumberLong(10000),
"numCatchUpOps" : NumberLong(0),
...
```

```
...
"newTermStartDate" : ISODate("2022-05-22T14:40:04.580Z"),
"wMajorityWriteAvailabilityDate" : ISODate("2022-05-22T14:40:05.990Z")
},
"members" : [
{
"_id" : 0,
"name" : "localhost:57017",
"health" : 1,
"state" : 1,
"stateStr" : "PRIMARY",
"uptime" : 488,
"optime" : {
"ts" : Timestamp(1653230724, 1),
"t" : NumberLong(1)
},
"optimeDate" : ISODate("2022-05-22T14:45:24Z"),
"lastAppliedWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
...
```

```
"lastDurableWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"syncSourceHost" : "",
"syncSourceId" : -1,
"infoMessage" : "",
"electionTime" : Timestamp(1653230404, 1),
"electionDate" : ISODate("2022-05-22T14:40:04Z"),
"configVersion" : 1,
"configTerm" : 1,
"self" : true,
"lastHeartbeatMessage" : ""
},
{
"_id" : 1,
"name" : "localhost:57018",
"health" : 1,
"state" : 2,
"stateStr" : "SECONDARY",
```

```
...
"uptime" : 331,
"optime" : {
"ts" : Timestamp(1653230723, 1),
"t" : NumberLong(1)
},
"optimeDurable" : {
"ts" : Timestamp(1653230723, 1),
"t" : NumberLong(1)
},
"optimeDate" : ISODate("2022-05-22T14:45:23Z"),
"optimeDurableDate" : ISODate("2022-05-22T14:45:23Z"),
"lastAppliedWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"lastHeartbeat" : ISODate("2022-05-22T14:45:24.482Z"),
"lastHeartbeatRecv" : ISODate("2022-05-22T14:45:23.545Z"),
"pingMs" : NumberLong(0),
"lastHeartbeatMessage" : "",
...
```

```
...
"syncSourceHost" : "localhost:57017",
"syncSourceId" : 0,
"infoMessage" : "",
"configVersion" : 1,
"configTerm" : 1
},
{
"_id" : 2,
"name" : "localhost:57019",
"health" : 1,
"state" : 2,
"stateStr" : "SECONDARY",
"uptime" : 331,
"optime" : {
"ts" : Timestamp(1653230723, 1),
"t" : NumberLong(1)
},
...
```

```
...
"optimeDurable" : {
"ts" : Timestamp(1653230723, 1),
"t" : NumberLong(1)
},
"optimeDate" : ISODate("2022-05-22T14:45:23Z"),
"optimeDurableDate" : ISODate("2022-05-22T14:45:23Z"),
"lastAppliedWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"lastDurableWallTime" : ISODate("2022-05-22T14:45:24.926Z"),
"lastHeartbeat" : ISODate("2022-05-22T14:45:24.481Z"),
"lastHeartbeatRecv" : ISODate("2022-05-22T14:45:23.520Z"),
"pingMs" : NumberLong(0),
"lastHeartbeatMessage" : "",
"syncSourceHost" : "localhost:57017",
"syncSourceId" : 0,
"infoMessage" : "",
"configVersion" : 1,
"configTerm" : 1
}
],
...
```

```
"ok" : 1,
"$gleStats" : {
"lastOpTime" : Timestamp(1653230393, 1),
"electionId" : ObjectId("7fffffff0000000000000001")
},
"lastCommittedOpTime" : Timestamp(1653230724, 1),
"$clusterTime" : {
"clusterTime" : Timestamp(1653230724, 1),
"signature" : {
"hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
"keyId" : NumberLong(0)
}
}, "operationTime" : Timestamp(1653230724, 1)
}
csReplSet:PRIMARY>
```

Paso 3

Se inician dos servidores de fragmentación en un conjunto de réplicas y se habilita la réplica entre ellos.

Shard 0

```
mkdir \data\shard0\rs0 \data\shard0\rs1 \data\shard0\rs2

mongod --replSets0 --dbpath \data\shard0\rs0 --port37017 --shardsvr
mongod --replSets0 --dbpath \data\shard0\rs1 --port37018 --shardsvr
mongod --replSets0 --dbpath \data\shard0\rs2 --port37019 --shardsvr
```

Salida en consola:

```
Cmder

E:\
λ mkdir \data\shard0\rs0 \data\shard0\rs1 \data\shard0\rs2

E:\
λ

E:\
λ mongod --replSet s0 --dbpath \data\shard0\rs0 --port 37017 --shardsvr
{"t":{"$date":"2022-05-22T08:00:20.002-07:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T08:00:20.003-07:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0

E:\
λ mongod --replSet s0 --dbpath \data\shard0\rs1 --port 37018 --shardsvr
{"t":{"$date":"2022-05-22T12:00:25.767-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificat
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"ma
WireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:00:25.770-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS

E:\
λ mongod --replSet s0 --dbpath \data\shard0\rs2 --port 37019 --shardsvr
{"t":{"$date":"2022-05-22T08:00:36.623-07:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificati
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"max
WireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T08:00:36.624-07:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS

cmd.exe   mongod.exe   mongod.exe   mongod.exe   Search
```

Shard 1

```
mkdir \data\shard1\rs0 \data\shard1\rs1 \data\shard1\rs2

mongod --replSets1 --dbpath \data\shard1\rs0 --port47017 --shardsvr
mongod --replSets1 --dbpath \data\shard1\rs1 --port47018 --shardsvr
mongod --replSets1 --dbpath \data\shard1\rs2 --port47019 --shardsvr
```

Salida en consola:

```
Cmder
E:\
λ mkdir \data\shard1\rs0 \data\shard1\rs1 \data\shard1\rs2

E:\
λ |

E:\
λ mongod --replSet s1 --dbpath \data\shard1\rs0 --port 47017 --shardsvr
{"t":{"$date":"2022-05-22T12:12:40.599-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:12:40.601-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0

E:\
λ mongod --replSet s1 --dbpath \data\shard1\rs1 --port 47018 --shardsvr
{"t":{"$date":"2022-05-22T12:12:52.537-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificati
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"max
WireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}

E:\
λ mongod --replSet s1 --dbpath \data\shard1\rs2 --port 47019 --shardsvr
{"t":{"$date":"2022-05-22T12:13:08.271-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:13:08.274-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS
cmd.exe  mongod.exe  mongod.exe  mongod.exe  Search
```

NOTA: MongoDB inicializa los primeros servidores de fragmentación como principal. Para mover el uso del servidor de fragmentación principal se puede utilizar el método **`movePrimary`**.

Paso 4

Se inicializa el conjunto de réplicas en los dos servidores fragmentados:

```
mongo --port37017

config = { _id:"s0", members: [
    {_id:0, host:'localhost:37017'},
    {_id:1, host:'localhost:37018'},
    {_id:2, host:'localhost:37019'}
]};
rs.initiate(config)
rs.status()
```

```
mongo --port47017

config = { _id:"s1", members: [
    {_id:0, host:'localhost:47017'},
    {_id:1, host:'localhost:47018'},
    {_id:2, host:'localhost:47019'}
]};
rs.initiate(config)
rs.status()
```

Después de ejecutar dichos comandos en consola se tendrá:

```
Cmder

s0:PRIMARY> 

E:\
λ mongod --replSet s0 --dbpath \data\shard0\rs0 --port 37017 --shardsvr
{"t":{"$date":"2022-05-22T08:00:20.002-07:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T08:00:20.003-07:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0
, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-22T08:00:20.005-07:00"},"s":"W",  "c":"ASIO",     "id":22601,   "ctx":"main","msg":"No TransportLayer configured

E:\
λ mongod --replSet s0 --dbpath \data\shard0\rs1 --port 37018 --shardsvr
{"t":{"$date":"2022-05-22T12:00:25.767-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificati
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"max
WireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:00:25.770-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS
1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}

E:\
λ mongod --replSet s0 --dbpath \data\shard0\rs2 --port 37019 --shardsvr
{"t":{"$date":"2022-05-22T08:00:36.623-07:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificati
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"max
WireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T08:00:36.624-07:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS

mongo.exe   mongod.exe   mongod.exe   mongod.exe
```

```
Cmder

s1:PRIMARY>

E:\
λ mongod --replSet s1 --dbpath \data\shard1\rs0 --port 47017 --shardsvr
{"t":{"$date":"2022-05-22T12:12:40.599-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:12:40.601-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automatically disabling TLS 1.0
, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}

E:\
λ mongod --replSet s1 --dbpath \data\shard1\rs1 --port 47018 --shardsvr
{"t":{"$date":"2022-05-22T12:12:52.537-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specificati
on","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"max
WireVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:12:53.070-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS
1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}

E:\
λ mongod --replSet s1 --dbpath \data\shard1\rs2 --port 47019 --shardsvr
{"t":{"$date":"2022-05-22T12:13:08.271-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification"
,"attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incomingInternalClient":{"minWireVersion":0,"maxWir
eVersion":13},"outgoing":{"minWireVersion":0,"maxWireVersion":13},"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:13:08.274-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"main","msg":"Automatically disabling TLS

mongo.exe   mongod.exe   mongod.exe   mongod.exe   Search
```

Paso 5

Se inician **`mongos`** para el conjunto fragmentado. Se crea una instancia y se indican los nodos que forman parte del **`config server`**.

```
mongos --configdbcsReplSet/localhost:57017,localhost:57018,localhost:57019
```

Salida en consola:

```
Cmder

E:\
λ mongos --configdb csReplSet/localhost:57017,localhost:57018,localhost:57019
{"t":{"$date":"2022-05-22T12:31:34.879-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initiali
zed wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":13},"incom
ingInternalClient":{"minWireVersion":0,"maxWireVersion":13},"outgoing":{"minWireVersion":13,"maxWireVersion":13}
,"isInternalClient":true}}}
{"t":{"$date":"2022-05-22T12:31:35.299-03:00"},"s":"I",  "c":"CONTROL",  "id":23285,   "ctx":"-","msg":"Automati
cally disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2022-05-22T12:31:35.310-03:00"},"s":"W",  "c":"ASIO",     "id":22601,   "ctx":"main","msg":"No Tr
ansportLayer configured during NetworkInterface startup"}
{"t":{"$date":"2022-05-22T12:31:35.311-03:00"},"s":"I",  "c":"NETWORK",  "id":4648602, "ctx":"main","msg":"Impli
cit TCP FastOpen in use."}
{"t":{"$date":"2022-05-22T12:31:35.312-03:00"},"s":"I",  "c":"HEALTH",   "id":5936503, "ctx":"main","msg":"Fault
 manager changed state ","attr":{"state":"StartupCheck"}}
{"t":{"$date":"2022-05-22T12:31:35.312-03:00"},"s":"W",  "c":"CONTROL",  "id":22120,   "ctx":"main","msg":"Acces
s control is not enabled for the database. Read and write access to data and configuration is unrestricted","tag
s":["startupWarnings"]}
{"t":{"$date":"2022-05-22T12:31:35.313-03:00"},"s":"W",  "c":"CONTROL",  "id":22140,   "ctx":"main","msg":"This
server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --b
ind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to
all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning",
"tags":["startupWarnings"]}
{"t":{"$date":"2022-05-22T12:31:35.314-03:00"},"s":"I",  "c":"CONTROL",  "id":23403,   "ctx":"mongosMain","msg":
"Build Info","attr":{"buildInfo":{"version":"5.0.8","gitVersion":"c87e1c23421bf79614baf500fda6622bd90f674e","mod
ules":[],"allocator":"tcmalloc","environment":{"distmod":"windows","distarch":"x86_64","target_arch":"x86_64"}}}
}

mongos.exe
Search
```

Paso 6

Se conecta el servidor de ruta **`mongo`** (por defecto se establece en el puerto 27017)

```
mongo
```

Salida:

```
Cmder

E:\
λ mongos --configdb csReplSet/localhost:57017,localhost:57018,localhost:57019

λ mongo
MongoDB shell version v5.0.8
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("0c1dd0f6-8637-4711-9b02-cb6b7d05391d") }
MongoDB server version: 5.0.8
================
Warning: the "mongo" shell has been superseded by "mongosh",
which delivers improved usability and compatibility.The "mongo" shell has been deprecated and will be removed in

an upcoming release.
For installation instructions, see
https://docs.mongodb.com/mongodb-shell/install/
================
---
The server generated these startup warnings when booting:
        2022-05-22T12:31:35.312-03:00: Access control is not enabled for the database. Read and write access to
data and configuration is unrestricted
        2022-05-22T12:31:35.313-03:00: This server is bound to localhost. Remote systems will be unable to conne
ct to this server. Start the server with --bind_ip <address> to specify which IP addresses it should serve respo
nses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with -
-bind_ip 127.0.0.1 to disable this warning
---
mongos>

mongos.exe   mongo.exe   Search
```

Se selecciona la base admin y se registran los dos **shards** creados en nuestra instancia de mongos, agregándole los servidores de fragmentación:

```
useadmin
sh.addShard('s0/localhost:37017')
sh.addShard('s1/localhost:47017')
```

Salida:

```
Cmder

E:\
λ mongos --configdb csReplSet/localhost:57017,localhost:57018,localhost:57019

mongos> use admin
switched to db admin
mongos> sh.addShard('s0/localhost:37017')
{
        "shardAdded" : "s0",
        "ok" : 1,
        "$clusterTime" : {
                "clusterTime" : Timestamp(1653233786, 5),
                "signature" : {
                        "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
                        "keyId" : NumberLong(0)
                }
        },
        "operationTime" : Timestamp(1653233786, 5)
}
mongos> sh.addShard('s1/localhost:47017')
{
        "shardAdded" : "s1",
        "ok" : 1,
        "$clusterTime" : {
                "clusterTime" : Timestamp(1653233796, 3),
                "signature" : {
                        "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
                        "keyId" : NumberLong(0)
                }
        },
        "operationTime" : Timestamp(1653233796, 3)
}
mongos>

mongos.exe   mongo.exe   Search
```

Luego, se puede consultar el *status*:

```
sh.status()
```

Salida:

```
Cmder
λ mongos --configdb csReplSet/localhost:57017,localhost:57018,localhost:57019

mongos> sh.status()
--- Sharding Status ---
  sharding version: {
        "_id" : 1,
        "minCompatibleVersion" : 5,
        "currentVersion" : 6,
        "clusterId" : ObjectId("628a4b4521141eaa5a661f6a")
  }
  shards:
        {  "_id" : "s0",  "host" : "s0/localhost:37017,localhost:37018,localhost:37019",  "state" : 1,  "topologyTime" : Timestamp(1653233786, 2) }
        {  "_id" : "s1",  "host" : "s1/localhost:47017,localhost:47018,localhost:47019",  "state" : 1,  "topologyTime" : Timestamp(1653233796, 1) }
  active mongoses:
        "5.0.8" : 1
  autosplit:
        Currently enabled: yes
  balancer:
        Currently enabled: yes
        Currently running: no
        Failed balancer rounds in last 5 attempts: 0
        Migration results for the last 24 hours:
                28 : Success
  databases:
        {  "_id" : "config",  "primary" : "config",  "partitioned" : true }
                config.system.sessions
                        shard key: { "_id" : 1 }
                        unique: false
                        balancing: true
                        chunks:
                                s0      996
                                s1      28
                        too many chunks to print, use verbose if you want to force print
mongos.exe   mongo.exe
```

Paso 7

Luego, se habilita la fragmentación en la base de datos destino (**`test`**)

```
usetest
sh.enableSharding('test')
```

Salida:

```
Cmder
λ mongos --configdb csReplSet/localhost:57017,localhost:57018,localhost:57019

mongos> use test
switched to db test
mongos> sh.enableSharding('test')
{
        "ok" : 1,
        "$clusterTime" : {
                "clusterTime" : Timestamp(1653234203, 3),
                "signature" : {
                        "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
                        "keyId" : NumberLong(0)
                }
        },
        "operationTime" : Timestamp(1653234203, 2)
}
mongos>

mongos.exe   mongo.exe                                   Search
```

Con este comando se verifica el status del *Sharding*

```
sh.status()
```

Salida:

```
Cmder
λ mongos --configdb csReplSet/localhost:57017,localhost:57018,localhost:57019
{"t":{"$date":"2022-05-22T12:31:34.879-03:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification","attr":{"spec":{"incomingExterna

mongos> sh.status()
--- Sharding Status ---
  sharding version: {
        "_id" : 1,
        "minCompatibleVersion" : 5,
        "currentVersion" : 6,
        "clusterId" : ObjectId("628a4b4521141eaa5a661f6a")
  }
  shards:
        {  "_id" : "s0",  "host" : "s0/localhost:37017,localhost:37018,localhost:37019",  "state" : 1,  "topologyTime" : Timestamp(1653233786, 2) }
        {  "_id" : "s1",  "host" : "s1/localhost:47017,localhost:47018,localhost:47019",  "state" : 1,  "topologyTime" : Timestamp(1653233796, 1) }
  active mongoses:
        "5.0.8" : 1
  autosplit:
        Currently enabled: yes
  balancer:
        Currently enabled: yes
        Currently running: no
        Failed balancer rounds in last 5 attempts: 0
        Migration results for the last 24 hours:
                219 : Success
  databases:
        {  "_id" : "config",  "primary" : "config",  "partitioned" : true }
                config.system.sessions
                        shard key: { "_id" : 1 }
                        unique: false
                        balancing: true
                        chunks:
                                s0      805
                                s1      219
                        too many chunks to print, use verbose if you want to force print
        {  "_id" : "test",  "primary" : "s1",  "partitioned" : true,  "version" : {  "uuid" : UUID("c4bb270e-b94d-4f25-922a-5338add673e9"),  "timestamp" : Timestamp(165
3234202, 11),  "lastMod" : 1 } }
mongos.exe   mongo.exe                                                                    Search
```

Paso 8

Sobre la base de datos **`test`** se crea la colección **`usuarios`**. Sobre ella, se aplicará el Sharding, insertando 10000 registros para que se pueda llevar a cabo la distribución.

```
for(vari=1; i<=10000; i++) {
    db.usuarios.insert({usuario_id:i, nombre:"nombre_"+i})
}
```

Para fragmentar la colección, se requiere de una clave de fragmentación (descripta más adelante, con detalle), en este caso será `usuario_id`.

Se deberá crear un índice sobre ella, ya que es condición necesaria que exista uno cuyo prefijo sea la *shared key* para poder llevar a cabo la distribución de los datos a lo largo de los shards:

```
db.usuarios.createIndex({usuario_id:1})
```

Salida en consola:

```
Cmder
"ordered":true,"lsid":{"id":{"$uuid":"363ee16f-5e99-4dc8-8216-d3500dd56121"}},"$clusterTime":{"clusterTime":{"$timestamp":{"t":1653236145,"i":17}},"si

mongos> for(var i=1; i<=10000; i++) {     db.usuarios.insert({usuario_id:i, nombre:"nombre_"+i}) }
WriteResult({ "nInserted" : 1 })
mongos> db.usuarios.count()
10000
mongos> db.usuarios.findOne()
{
        "_id" : ObjectId("628a5f9f81398830fef5c3e5"),
        "usuario_id" : 1,
        "nombre" : "nombre_1"
}
mongos> db.usuarios.createIndex({usuario_id:1})
{
        "raw" : {
                "s1/localhost:47017,localhost:47018,localhost:47019" : {
                        "numIndexesBefore" : 1,
                        "numIndexesAfter" : 2,
                        "createdCollectionAutomatically" : false,
                        "commitQuorum" : "votingMembers",
                        "ok" : 1
                }
        },
        "ok" : 1,
        "$clusterTime" : {
                "clusterTime" : Timestamp(1653236833, 5),
                "signature" : {
                        "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
                        "keyId" : NumberLong(0)
                }
        },
        "operationTime" : Timestamp(1653236833, 5)
}
mongos.exe    mongo.exe
```

Paso 9

En este momento, los datos se encuentran dentro del *shard* primario ya que no se ha habilitado todavía el *sharding* sobre esta colección.

```
sh.shardCollection('mibase.usuarios', {usuario_id:1})
```

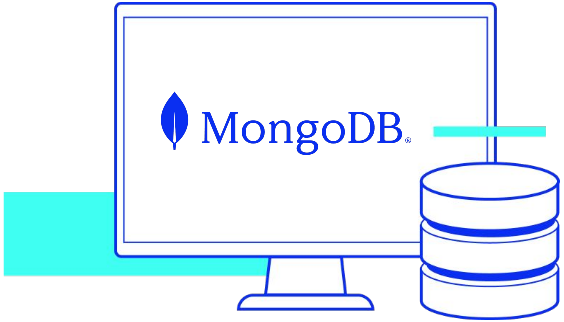
Salida en consola:

```
Cmder
{"t":{"$date":"2022-05-22T13:17:18.644-03:00"},"s":"I",  "c":"COMMAND",  "id":51803,   "ctx":"conn47","

mongos> sh.shardCollection('test.usuarios', {usuario_id:1})
{
        "collectionsharded" : "test.usuarios",
        "ok" : 1,
        "$clusterTime" : {
                "clusterTime" : Timestamp(1653238481, 12),
                "signature" : {
                        "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAA="),
                        "keyId" : NumberLong(0)
                }
        },
        "operationTime" : Timestamp(1653238481, 8)
}
mongos>
```

Con este comando se verifica:

```
db.usuarios.getShardDistribution()
```



```
efaultWriteConcern":{"w":"majority","wtimeout":0},"defaultWriteConcernSource":"implicit","defaultReadCo

mongos> db.usuarios.getShardDistribution()

Shard s1 at s1/localhost:47017,localhost:47018,localhost:47019
 data : 643KiB docs : 10000 chunks : 1
 estimated data per chunk : 643KiB
 estimated docs per chunk : 10000

Totals
 data : 643KiB docs : 10000 chunks : 1
 Shard s1 contains 100% data, 100% docs in cluster, avg obj size on shard : 65B

mongos>
```

Para favorecer la distribución de los datos, se establece un tamaño de *chunk* de 1MB ya que por defecto es de 64MB. Dicho tamaño determina la cantidad de divisiones y migraciones de *chunks* que se van a realizar.

```
useconfig
db.settings.save({_id:"chunksize", value:1})    //value está en MB
```

Salida:

```
che-14","msg":"Refreshed cached collection","attr":{"namespace":"config.system.sessions","lookupSinceVe
mongos> use config
switched to db config
mongos> db.settings.save({_id:"chunksize", value:1})
WriteResult({ "nMatched" : 0, "nUpserted" : 1, "nModified" : 0, "_id" : "chunksize" })
mongos> use test
switched to db test
mongos>
```

Paso 10

Se insertan 90000 documentos más en la colección usuarios:

```
for(vari=10001; i<=100000; i++) {
    db.usuarios.insert({usuario_id:i, nombre:"nombre_"+i})
}
```

Salida en consola:

```
Cmder
"t":1653233839,"i":6}}},"newVersion":{"chunkVersion":{"0":{"$timestamp":{"t":513,"i":1}},"1":{"$oid":"6
mongos> for(var i=10001; i<=100000; i++) {     db.usuarios.insert({usuario_id:i, nombre:"nombre_"+i}) }

WriteResult({ "nInserted" : 1 })
mongos> db.usuarios.count()
100000
mongos>
```

El último paso es **verificar el estado de la fragmentación.**
En la siguiente presentación se detalla esta operación.

¡Sigamos trabajando!